

Original software publication

Atlantic cod growth model: Open source Python package for numerical growth experiments

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ABSTRACT

In this article we introduce an open source tool that is used to study growth of Atlantic cod (*Gadus morhua*), an economically important fish species, in different temperature environments. It is a mechanistic physiology-based growth model that simulates growth in controlled laboratory experiments as well as in an open sea environment. The model can be used by universities, industrial, and research institutes as a tool for testing research hypotheses, and studying temperature-dependent growth in fishes.

Code metadata

Current code version	https://github.com/ElsevierSoftwareX/SOFTX-D-24-00617
Permanent link to code/repository used for this code version	https://github.com/n-sokolova/growth-model-atlantic-cod
Permanent link to Reproducible Capsule	https://zenodo.org/records/14395094
Legal Code License	MIT License
Code versioning system used	git
Software code languages, tools, and services used	Python 3, JavaScript
Compilation requirements, operating environments & dependencies	tested under Windows, macOS Monterey 12.4. Preferable OS: Ubuntu 22.04.3 LTS 64-bit. Dep.: Python >=3.9,<3.11[?]; xarray 2022.3.0; tomli 2.0.1; netCDF4 1.5.8; dask 2022.5.2
If available Link to developer documentation/manual	https://github.com/n-sokolova/growth-model-atlantic-cod
Support email for questions	nadezda.v.sokolova@gmail.com

1. Motivation and significance

Growth is one of the main properties of life as well as a key physiological and ecological trait [1]. Growth of marine organisms is influenced by many interacting factors that differ across species and vary with time scales. Laboratory findings show that the growth rate of an organism directly responds to temperature changes, espe-

cially in ectotherms — organisms whose body temperature depends on surrounding water temperatures (e.g. fishes). Studies show that fish growth rates will change under global ocean warming, in particular that maximum fish size may decrease if ocean temperatures continue rising [2]. Here, we present a model which enables to simulate the growth of Atlantic cod, an economically important fish species, for various scenarios of ocean warming considering a wide range of spatial and

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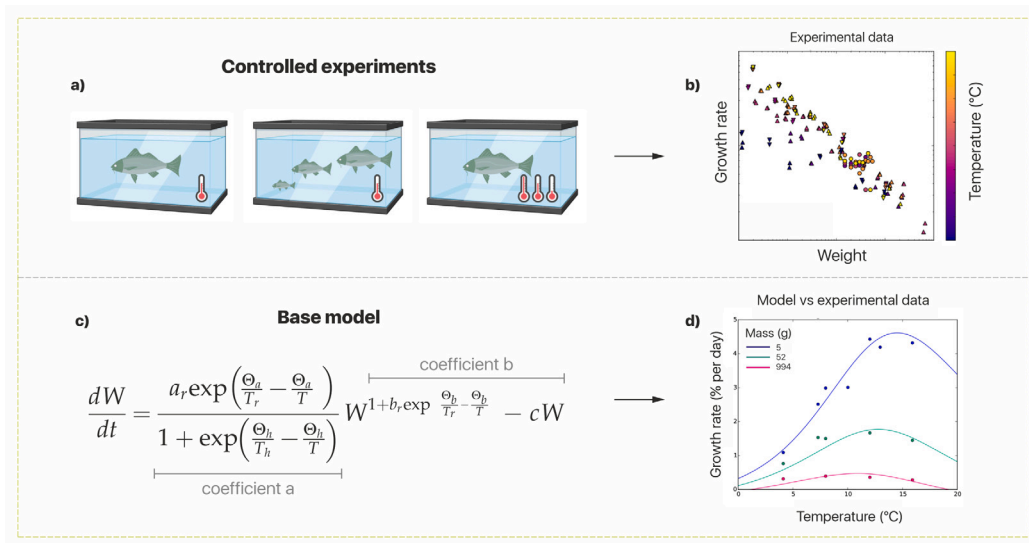


Fig. 1. Conceptual figure describing the workflow: obtaining growth data from laboratory experiments, using these data to derive the mathematical equation for growth and visualizing the model behavior with different input parameters.

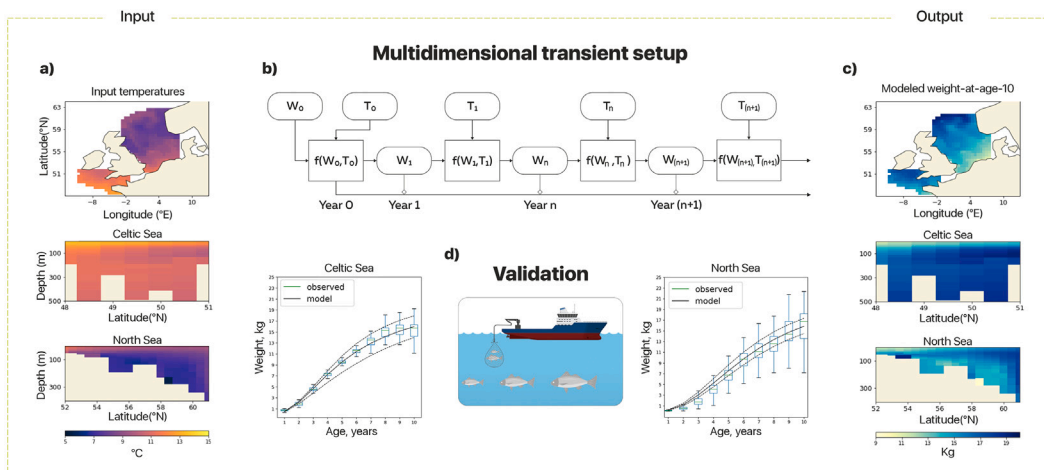


Fig. 2. Concept figure describing multidimensional transient setup. The model uses a multidimensional (lon, lat, depth, time) input (a) and calculates growth rates for each year at a daily time step (b). The output data have the same resolution as the input data (c). Median weight-at-age values are calculated for each age class (d - boxplots).

temporal scales. By definition, a “growth model” is any quantitative description of growth from theoretically informed mathematical equations to purely empirical or statistical models [3]. The most common approaches to simulate fish growth were developed by von Bertalanffy and Gompertz [3–5]. Some of those models, written in the programming language R, have been already introduced on open source platforms (e.g. [6,7]). Our source code is written in Python and available on GitLab and Github [8]. Our software includes the growth model equations derived by Butzin and Pörtner (2016) [9] and an extended version of the growth model [10] both validated and published.

1.1. Conceptual framework

A bio-physical mechanistic background of the model roots in controlled laboratory experiments for Icelandic population of Atlantic cod (the data obtained from [11,12]) where fish growth was measured in fish tanks with various constant temperature settings ($T = [2, 4, 7, 8, 10, 12, 13, 16]^\circ\text{C}$) for any given individual weight from 0.001 to 8.7 kg (Fig. 1a). Laboratory experiments are characterized by the regimes of constant temperature, salinity and oxygen. The time frame spans hours to months. Experimental output includes growth rates measured as increments in body weight in relation to temperature

(Fig. 1b). Based on these data a mathematical equation was derived that includes a temperature dependent nonlinear reaction model with growth inhibition at 12°C (coefficient a) and temperature dependent allometric scaling exponent b (Fig. 1c; see Listing 4, 5 for Python code). As the model equations were derived from experimental data, the base model setup mimics the results of controlled laboratory experiments by simulating temperature-dependent growth for individuals of different sizes (Fig. 1d). The base model could be considered as a generalized von Bertalanffy growth model. Details on calibration and procedure of model development are described in [9]. Key model assumptions: growth rates scale with body size — allometric growth [13]; immediate response of organisms to temperature changes [9]; the growth response is accompanied by unlimited food supply (based on growth data from [11,12]). In order to simulate growth in the natural environment we incorporated multi-dimensional temperature datasets (SODA) into the base model setup (Fig. 2). The multi-dimensional transient setup simulates continuous growth of cod over its life-cycle for any given time period in any location of the North Atlantic. The model output (weight-at-age) is in the same format and provides the same temporal resolution as field data reported by the International Council for the Exploration of the Sea [14–16], which allows direct comparison of modeled and observed values.

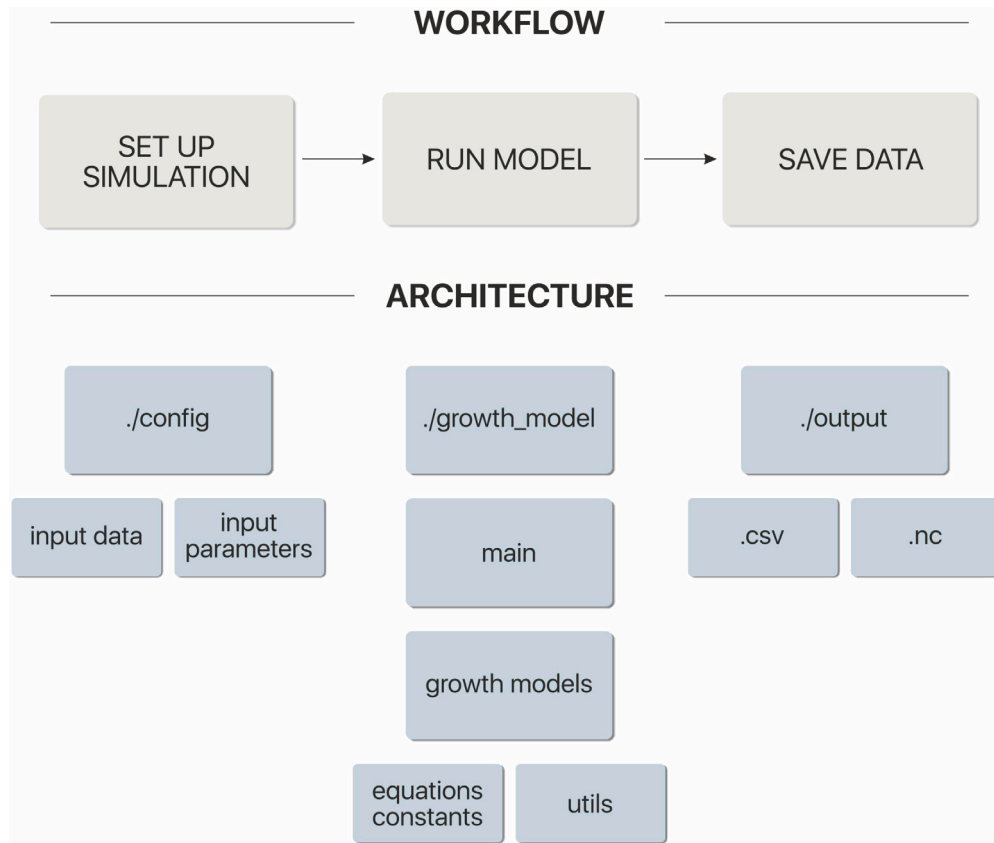


Fig. 3. Workflow and package architecture. The workflow includes model setup, growth model simulation, and output data saving. A user sets up a simulation in a config file by specifying input temperature data and configuration parameters (e.g. max age in a life cycle, initial weight of individual) for a specific growth model. Growth simulation at runtime involves numerical calculations using growth model equations and constants. The growth model scripts are linked to the main executable file. Input and output data processing is supported by functions in utils. The outputs are saved in a directory specified by a user in a command line (.csv or .netcdf formats).

2. Software description

2.1. Software architecture

The growth model repository structure is shown in Fig. 3. It is implemented as a Python package with a simple user setup. The package structure is based on two central components: the source code written in Python (version 3.9, [17]) and user configuration files in the .toml format. The source code architecture comprises the following elements: growth model equations and constants (Listing 4); main executable file (Fig. 3: main) that calls a certain function based on a given growth model configuration (Tables A.1 and A.2); growth model scripts (Fig. 3: growth models); supporting functions (Fig. 3: utils). The main executable file incorporates parameters from the config file and arguments from the growth model functions.

2.2. Software functionalities and system requirements

To set up a simulation a user chooses one of the available models. There are five model configurations a user can apply (Tables A.1 and A.2). In a given config file a user defines input parameters of the model — initial and boundary settings — and runs the model (Listing 1). The executable “cod-growth-model” is called via the command line. Below is an example of the command line arguments that a user should give in to get the provided output for the base model.

```
Syntax:
$ cod-growth-model [CONFIG FILE] [OUTPUT
FILE or FOLDER]
```

```
Example:
$ cod-growth-model ./config/base.toml ./
output/base_model.csv
```

```
Output:
Initial weight: 1000 g
Input temperature: 10 °C
Relative growth rate: 0.0046543 per day
Calculated weight: 1004.65432 g
```

Listing 1: Running the model (base configuration)

Based on a research question a user can choose any setting: from the base model setup to the complex multidimensional configuration (the growth model consists of five configurations described in Tables A.1 and A.2). Depending on the model configuration, a config file should provide information about input temperature values, geographical boundaries, biological parameters (e.g. max age in a life cycle, initial weight of individual), input and output directories. Parameters in a config file have various data types: integers or floating point numbers, lists, and strings.

3. Illustrative examples

Here we provide the workflow for multidimensional model (multi_dim). A running command is shown in Listing 2, a config file – in Listing 3 and output data structure – in Fig. 4. The default con-

figuration models growth of Celtic Sea cod over a 5-year life cycle for years 1980–1989. The model uses the Python xarray library [18] to create, process and save variables (a_3d, b_3d, growth_rates_3d, weight_3d, weight_max) in multidimensional .netcdf files. The multi_dim model inputs and outputs are represented in the .netcdf format (Unidata's Network Common Data Form). This is a portable binary data format that supports creation, access, and sharing of array-oriented metadata. The default input dataset is stored in a repository folder named input_data. User can run the model with an alternative dataset and provide the path to the input data in a related multi_dim config file. The output file names are created automatically based on a given variable, experiment name, region name, and simulation year.

```
$ cod-growth-model ./config/multi_dim.toml
./output/
```

Listing 2: Command line call executing the multidimensional model

```
output/SODA/CelticSea/
├── a_3d
│   ├── 1981_a_3d_age1.nc
│   ├── 1982_a_3d_age2.nc
│   ├── ...
│   └── 1985_a_3d_age5.nc
├── b_3d
│   ├── 1981_b_3d_age1.nc
│   ├── ...
│   └── 1985_b_3d_age5.nc
├── growth_rates_3d
│   ├── 1981_growth_rates_3d_age1.nc
│   ├── ...
├── weight_3d
│   ├── 1981_weight_3d_age1.nc
│   ├── ...
│   └── 1985_weight_3d_age5.nc
└── weight_max
    ├── 1981_weight_max_age1.nc
    ├── ...
    └── 1985_weight_max_age5.nc
```

Fig. 4. Output files tree. The output variables are 3-dimensional. They represent a and b coefficients, growth rates, weight-at-age and maximum asymptotic weight from [9].

```
model = "multi_dim"
'''
    Define coordinates (longitudes, latitudes)
    of your region
    These coordinates will be used for
    extracting input
    temperature data from netcdf file
    and saving output weight-at-age data to
    netcdf
'''

lat, lon = [
    47.25, 47.75, 48.25, 48.75, 49.25,
    49.75, 50.25,
    50.75, 51.25, 51.75
], [
    -11.75, -11.25, -10.75, -10.25, -9.75,
    -9.25, -8.75,
    -8.25, -7.75, -7.25, -6.75, -6.25,
    -5.75, -5.25,
    -4.75, -4.25, -3.75, -3.25, -2.75,
    -2.25, -1.75, -1.25
]

# Define same depth levels as in your netcdf
file
depth = [
    30, 40, 50, 60, 70, 80, 90, 100,
    115,
    135, 160, 190, 230, 280, 340, 410, 490,
    580
]

# Number of years in one life cycle of an
individual
max_age = 5

'''
    Input dataset parameters
    Range of years in the input temperature
    dataset
    Initial year should be one year less than
    starting
    year in the dataset
'''
initial_year, final_year = 1980, 1989

# Referenced output folders
exp_name, region, output = "SODA", "
CelticSea", "./output/"
```

Listing 3: Example config file for the multidimensional model

Depending on the model configuration outputs are saved in the comma separated ASCII data format “.csv” (base, one_init_weight, one_temp_input, one_dim), or “.netcdf” which is a common data format used by climate model <https://www.unidata.ucar.edu/software/netcdf/> (multi_dim) formats. The growth model outputs are saved in an automatically created directory (./output) defined in a command line.

4. Impact

In this article we propose publicly available open source model code that could be used in order to facilitate model development and hypothesis testing in fish growth studies. Dimensionless versions (base, one_temp_input, one_init_weight) help testing experimentally derived growth parameters for Atlantic cod populations from

different regions. Further applications of the model could involve studies on the effects of physiological adaptation on fish growth [19]. Furthermore, the multidimensional version of the model (`multi_dim`) allows study growth on different spatial and temporal scales providing outputs that can be directly compared to field data [10].

The source code can be also used as a practical teaching tool. The model code was translated to JavaScript and published as an interactive web page introducing growth studies to undergraduate students (https://www.awi.de/fileadmin/user_upload/AWI/Forschung/Klimawissenschaft/Dynamik_des_Palaeoklimas/Growth_Model/index.html, see sample JavaScript code in Listing 7). The web page repository is also an open source project published on GitHub: <https://github.com/btschwertfeger/Growth-Model-Website>.

For further development and research the authors consider enhancing the model with weight and age information back-calculated based on fish otoliths [20].

CRedit authorship contribution statement

Nadezhda Sokolova: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Anja Rohner:** Writing – review & editing, Writing – original draft, Validation, Software. **Martin Butzin:** Writing – review & editing, Validation, Supervision, Software, Methodology, Conceptualization. **Hans-Otto Poertner:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Gerit Lohmann:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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- Paul Gierz (Alfred Wegener Institute, Helmholtz Centre for Polar and Marine Research, Bremerhaven, Germany) <https://orcid.org/0000-0002-4512-087X>

Appendix. Supplements

```
"""
Definitions of growth models constants
from Butzin and Poertner, 2016
"""

A_R = 8.660          # Uninhibited growth rate
                      # at reference temperature T_R (% d^-1 g^-1/b)
B_R = 0.3055         # Value of allometric
                      # exponent at reference temperature Tr
THETA_A = 18145      # Arrhenius temperature (K)
                      # for uninhibited reaction kinetics =
                      # 17871.85rC
THETA_B = 4258       # Arrhenius temperature (K)
                      # = 3984.85rC
THETA_H = 25234      # Arrhenius temperature (K)
                      # for inhibited reaction kinetics = 24960.85
                      # rC
T_R = 283            # Reference optimum
                      # temperature (K) = 9.85rC
T_H = 286            # Temperature for
                      # inhibitive processes (K) = 12.85rC
C_AVG = 0.291        # Independent of
                      # temperature and weight constant (%d^-1)
```

Listing 4: Python code for constants from Eq. 2 and 3 in Butzin and Poertner (2016).

```
"""
In this file functions for coefficients a
and b are defined (Eq. 2 and 3 in Butzin and
Poertner, 2016)
"""

from growth_model.constants import A_R, THETA_A,
    T_R, THETA_H, T_H, B_R, THETA_B, TO

from numpy import exp

def equation2(input_temp):
    """
    This function calculates parameter "a" (Eq.
    2 in Butzin and Poertner, 2016). This is a
    nonlinear master reaction model of
    poikilotherm development with growth
    inhibition at higher temperatures.
    @param input_temp: input temperature for
    growth rate calculation (°C)
    @return: a
    """
    temperature_kelvin = input_temp + TO
    # Calculate a
    a_numerator = A_R * exp(THETA_A / T_R -
        THETA_A / temperature_kelvin)
    a_denominator = 1 + exp(THETA_H / T_H -
        THETA_H / temperature_kelvin)
    a = a_numerator / a_denominator
    return a

def equation3(input_temp):
    """This function calculates an allometric
    scaling exponent "b"
    (Arrhenius equation, Eg. 3 in Butzin and
    Poertner, 2016)

    @param input_temp: input temperature for
    growth rate calculation (°C)
```

```

@return: b
"""
temperature_kelvin = input_temp + T0
# Calculate b
b = B_R * exp(THETA_B / T_R - THETA_B /
temperature_kelvin)
return b

```

Listing 5: Python code for coefficients a and b from Eq. 2 and 3 in Butzin and Poertner (2016).

```

def main():
    """
    Entrypoint of the command line interface for
    interacting with cod-growth-models

    @return: exit code
    """

    # Initialize logger
    logger_config = pathlib.Path.cwd().joinpath(
        'logger.conf')
    init_logger(logger_config)

    # Parse cli arguments
    args = _parse_args()
    settings = load_config(args.config_file)

    model = settings['model']
    output = pathlib.Path(args.output)

    models_output_dir = ['multi_dim']
    models_output_file = ['base', 'one_dim',
                          'one_weight', '
one_weight']

    if model in models_output_dir and output.
is_file():
        sys.exit(f'Error:
Output must be directory for models
{models_output_dir}')
    if model in models_output_file and output.
is_dir():
        sys.exit(f'Error:
Output must be a file for models
{models_output_file}')

    if model == 'base':
        logging.info
        (f'Calculating growth model with model "
base"...')
        try:
            base(temp=settings['temp'],
weight=settings['weight'],
output=output)
        except KeyError:
            sys.exit(f'Error:
Model-Settings for model "base" in
config-file
{args.config_file} are incomplete!')

    elif model == 'one_dim':
        logging.info
        (f'Calculating growth model with model "
one_dim"...')
        try:
            one_dim(min_temp=settings['min_temp'
],
max_temp=settings['max_temp'],
temp_step=settings['temp_step'],
weight=settings['weight'],

```

```

max_age=settings['max_age'],
output=output)
        except KeyError:
            sys.exit(f'Error: Model-Settings for
model "one_dim" in
config-file {args.config_file}' f'
are incomplete!')

    elif model == 'one_temp':
        logging.info
        (f'Calculating growth model with model "
one_temp"...')
        try:
            one_temp(temp=settings['temp'],
weight=settings['weight'],
output=output)
        except KeyError:
            sys.exit
            (f'Error: Model-Settings for model "
one_temp" in
config-file {args.config_file} are
incomplete!')

    elif model == 'one_weight':
        logging.info
        (f'Calculating growth model with model "
one_weight"..')
        try:
            one_weight(temp=settings['temp'],
weight=settings['weight'],
output=output)
        except KeyError:
            sys.exit(f'Error:
Model-Settings for model "one_weight
"
in config-file {args.config_file}
are incomplete!')

    elif model == 'multi_dim':
        logging.info
        (f'Calculating growth model with model "
multi_dim"...')
        try:
            multi_dim(lat=settings['lat'],
lon=settings['lon'], depth=settings[
'depth'],
max_age=settings['max_age'],
first_year=settings['first_year'],
final_year=settings['final_year'],
exp_name=settings['exp_name'],
region=settings['region'],
input_data=settings['input_data'],
output=output)
        except KeyError:
            sys.exit
            (f'Error:
model "multi_dim" expects parameter-
settings!')
        else:
            logging.critical(f'Error: model "{model
}" not available')
            sys.exit(1)

```

Listing 6: Sample code of mainpy that incorporates parameters from the config file and arguments from the growth model functions.

```

run_model = (
    data = null, parameters = null,
    initial_weight = null
) => {

```


Table A.1
Growth models and their configurations (Part 1).

Model/Config name	Description	Inputs	Outputs
base	A dimensionless model that calculates relative growth rate and final weight of one individual considering one constant temperature environment	Temperature: single value (integer), initial weight: single value (integer)	Relative growth rate: single value, final weight: single value, file format: csv
one_temp_input	A dimensionless model that calculates relative growth rate and final weight of several individuals considering one constant temperature environment	Temperature: single value (integer), initial weight: several values (list)	Relative growth rates: list, final weights: list, file format: csv
one_init_weight	A dimensionless model that calculates relative growth rate and final weight of one individuals considering several constant temperature environments	Temperature: several values (list), initial weight: one value (integer)	Relative growth rates: list, final weights: list, file format: csv

Table A.2
Growth models and their configurations (Part 2).

Model/Config name	Description	Inputs	Outputs
one_dim	A 1-dimensional model that calculates continuous growth of several individuals over a given time period under several constant temperature regimes (Fig. 2b in Butzin and Pörtner, 2016).	Temperature: minimum temperature (integer), maximum temperature (integer), temperature step (float); initial weight: single value (integer); maximum age of individual in years (integer)	Relative growth rates and final weights: table, file format: csv
multi_dim	A multi-dimensional transient model (updated setup from Sokolova et al. (2021). The model imitates monthly growth and calculates weight at a certain age (weight-at-age) using multidimensional ocean temperature data (space-, depth- and time-varying)	Temperature: multi-dimensional temperature dataset in netCDF format (minimum 10 years time series); temporal boundaries: first and last year in a dataset (integers); geographic boundaries: latitudes, longitudes, depths levels (lists); maximum age of individual in years (integer)	Annual multi-dimensional datasets: a_3d , b_3d , growth_rates_3d , weight_3d , weight_max ; file format: netCDF

```

data = (data === null) ?
  window.process_data_simple : data;
parameters = (parameters === null) ?
  window.default_input_simple :
parameters;
initial_weight = (initial_weight ===
null) ?
  parameters.initial_weight * 1000
  : parseFloat(initial_weight) * 1000;

let
  a_fit = data.a_fit,
  b_fit = data.b_fit,
  c_avg = data.c_avg,
  itemp = 0,
  dt = 1, // time step in days
  temperature_range = range(-2, 30.5 -
.5, .5);

/* weight at age (in g) for a given
temperature
&& initial weight in kg
*/

```

```

let weight_at_age = [...new Array(
  parameters.max_age
)].map(
  (e, i) => [...new Array(
    a_fit.length
  )].map(() => (i === 0) ?
initial_weight : 0)
);

// loop over temperatures, increasing
values by 0.5°C
temperature_range.forEach(() => {
  range(1, parameters.max_age - 1, 1).
forEach((age) => {
    let growth_rate = .01 * (
      a_fit[itemp] * Math.pow(
        weight_at_age[age - 1][
itemp], b_fit[itemp]
      ) - c_avg
    );
    weight_at_age[age][itemp] =
      weight_at_age[age - 1][itemp]
    * (

```

```

        1 + dt * growth_rate
    );
});
itemp++;
});

return {
    a_fit: a_fit, b_fit: b_fit,
    c_avg: c_avg,
    weight_at_age: weight_at_age,
    temperature_range: temperature_range
},
}
}

```

Listing 7: Growth Model in JavaScript

Data availability

Data is available on gitlab and PANGAEA.

Input temperature data:

- <https://doi.pangaea.de/10.1594/PANGAEA.943546>
 - https://codebase.helmholtz.cloud/awi_paleodyn/growth_model_atlantic_cod_inputs
- Weight data:
- https://codebase.helmholtz.cloud/awi_paleodyn/atlantic_cod_growth_data

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